

SNIGDHA GAYATHRI MANDAPATI

AI Engineer | LLM Systems, RAG Pipelines & Multi-Agent Orchestration

+91 8247310334 | snigdhaisme@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

AI Engineer specializing in **LLM-powered systems**, **agentic architectures**, and production-grade **RAG pipelines**, with hands-on depth across the full GenAI stack: **transformer** internals, **retrieval and vector search**, **fine-tuning** (LoRA/QLoRA), multi-agent orchestration with **LangChain/LangGraph**, and **MLOps** for model serving and evaluation. Proven track record shipping end-to-end AI systems: from **5-agent orchestration logic** and security-hardened RAG pipelines to **privacy-preserving hybrid ML architectures**. Combines strong software engineering fundamentals (Python, FastAPI, React, PostgreSQL) with a security- and evaluation-first mindset, consistently **reducing latency, hallucination risk, and model drift in deployed systems**.

SKILLS

LLM & Generative AI Engineering: Large Language Models, Transformer Architecture, Prompt Engineering, LLM Evaluation & Benchmarking, LLMops, Hallucination Mitigation, Generative AI, Context Engineering, Gemini API, Hugging Face Transformers

Agentic AI & Orchestration: LangChain, LangGraph, LlamaIndex, Multi-Agent Systems, Agent Memory, Tool Use / Function Calling, Model Context Protocol (MCP), Structured Output Generation, Agent Evaluation & Reliability Patterns

RAG & Retrieval: Retrieval-Augmented Generation, Vector Databases, Embedding Models (SentenceTransformers), Semantic Search, Chunking Strategies, Reranking, Grounding & Faithfulness Verification, RAGAS

Fine-Tuning & Model Development: LoRA / QLoRA, Supervised Fine-Tuning (SFT), Scikit-Learn, XGBoost, TensorFlow, PyTorch, Quantum ML (PennyLane), Model Evaluation (F1, ROC-AUC, RMSE)

ML, MLOps & Deployment: Machine Learning, Deep Learning, NLP, Data pre-processing, Data Analysis, ML pipelines, Docker, FastAPI, Flask REST APIs, CI/CD, Model Monitoring & Drift Detection, LLM Observability (LangSmith), Cost & Latency Optimization

Programming & Data: Python, JavaScript/TypeScript, SQL, Java, C++, PostgreSQL, MySQL, Next.js, React, Git, JWT/Auth (Clerk)

TECHNICAL EXPERIENCE

Machine Learning Intern | Cognifyz Technologies

Jan 2025 – Mar 2025

- Deployed 10+ supervised ML models (Scikit-Learn, XGBoost) as Dockerized Flask REST APIs, sustaining sub-200ms latency across 10K+ daily requests in production.
- Reduced production model drift by ~30% by designing and shipping automated monitoring and retraining pipelines, while achieving 95% validation accuracy on skewed real-world datasets.

Artificial Intelligence Intern | Technical Hub Pvt. Ltd.

May 2024 – Sep 2024

- Increased retrieval precision by 40% over the prior baseline by architecting multi-stage classification pipelines (PCA, ensemble methods) with iterative feature engineering.
- Cut inference pipeline processing time by 35% and accelerated model iteration cycles by 25% by profiling and re-engineering the ML pipeline, shortening the team's experiment-to-result loop.

PROJECTS

Agentic Placement RAG | Security-Hardened, Production-Grade RAG Pipeline

[Link](#)

- Delivered** a security-hardened Agentic RAG platform as measured by a **9-stage guardrail pipeline** and **≥90% test coverage**, by architecting a FastAPI + LangGraph backend with **hybrid BM25 + Gemini retrieval** and a defined production-scaling roadmap.
- Improved retrieval precision** and **reduced hallucinations** in a live RAG assistant as measured by cross-encoder re-ranking and **grounded-output verification**, by building **multi-hop query rewriting** and **SSE observability**, then shipping to production on Render.

Smart Shelf AI | Hybrid Quantum-Classical Recommendation Engine

[Link](#)

- Reduced** recommendation latency from a 1–2 minute cold start to sub-1-second on repeat queries by engineering a **hybrid classical quantum pipeline** (SentenceTransformers 384-dim embeddings + PennyLane 6-qubit similarity kernel) with a precomputed caching layer.
- Delivered** a **100% local-inference**, zero-external-dependency recommendation engine : including **27-class emotion detection** (GoEmotions) : served through a dedicated FastAPI backend for full data privacy.

NeuroPlan AI | Multi-Agent Adaptive Learning System

[Link](#)

- Architected** a **5-agent orchestration system** (Roadmap Creation, Fatigue Monitoring, Transfer Network, Mastery Radar, Planner) that autonomously generates, monitors, and re-optimizes **personalized learning roadmaps in real time**.
- Built** cross-agent coordination logic enabling the Planner Agent to arbitrate fatigue, mastery, and transfer signals and **dynamically re-sequence tasks**, on a **FastAPI + Next.js + PostgreSQL stack** with Gemini-powered generation and Clerk-secured access.

EDUCATION

B.Tech in Computer Science & Engineering (AI/ML) — Aditya University, Surampalem

2022 – 2026 | CGPA: 8.30

CERTIFICATIONS

Microsoft (**Azure AI Fundamentals**) | Google (Career Essentials in **Generative AI**) | IBM (**ML with Python, DL Fundamentals**, Data Science 101) | LangChain Academy (Introduction to **LangChain** – Python, Building **Reliable Agents**) | SimpliLearn (**LangChain for LLM App Development**) | ActiveLoop (**Retrieval Augmented Generation for Production** with LangChain & LlamaIndex) | Pearson (IT Specialist – Java)